



iCanStudy: Upgrading learning for millions using data and personalized feedback with Gemini

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GOOGLE CLOUD RESULTS

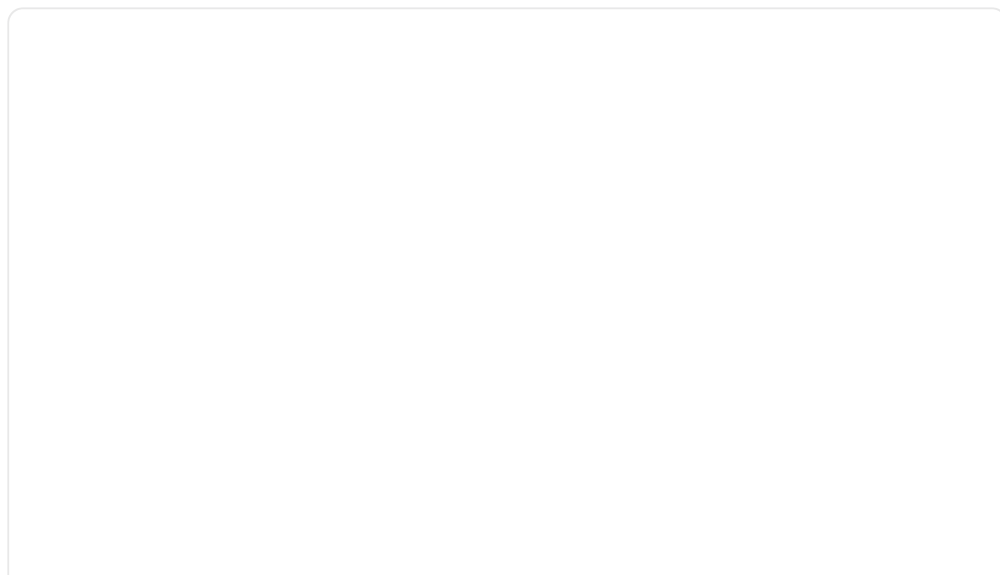
- ✓ Achieved a world-first in analyzing and providing feedback on learners' cognitive processes using 1,000+ variables at scale
- ✓ Feedback is received 288 times faster than from a human coach, enabling a high-frequency feedback loop
- ✓ Uncovering previously unattainable learning trends and patterns in educational research with Vertex AI and Google Cloud
- ✓ Increased capacity from thousands to 1M+ global users

With Google Cloud, iCanStudy has deployed a new Learning Experience Platform that can support more than one

million users, by using AI-powered services that improve the efficiency and effectiveness of learning by 10x.

Founded in late 2020, [iCanStudy](#) is an educational organization that applies research, practical testing, and best-practice training methodologies to teach individuals to learn more efficiently. The edtech business works directly with tens of thousands of learners ranging from adolescents to seasoned professionals from over 120 countries—as well as educators, institutions, and enterprises—by applying cognitive retraining programs that enhance memory, problem-solving, understanding, and deep processing skills. According to Michael Tsai, iCanStudy Chief Executive Officer, iCanStudy's methodologies deliver consistent, observable improvements across its learner cohort. Individuals spend just 20 minutes every third day to improve their ability to acquire, process, retain, and apply new knowledge and skills faster and deeper, like a seasoned subject matter expert.

iCanStudy's plan to deepen and accelerate learning emerged from its leader's belief that existing learning management systems could not support the teaching of complex cognitive skills. "Existing options missed steps, which led to unsuccessful learning outcomes in skills development, and no one has ever incorporated important pedagogical principles into existing learning management systems," said Tsai. "To bypass the limitations of conventional learning management systems, we decided to build our own proprietary Learning Experience Platform to create a learning experience with integrated features to optimize skills development." The business also wanted to leverage the opportunities AI presented to enhance learning and Google Cloud was the ideal partner for this journey.





Hyperscaling to one million plus users

Until early 2023, iCanStudy was running its website and traditional learning platform on a virtual server in a third-party cloud service. However, that server was unable to support growing demand and the business's roadmap, with its capacity limited to 5,000 concurrent active users. Without the ability to scale, the iCanStudy website was experiencing high latency or even crashing during periods of high demand, such as enrolment periods, affecting sales, revenue, and reputation. "We needed a solution that would allow us to hyperscale to one million plus active customers, with a modular architecture that enabled load balancing and could upscale or

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Archer Newton

downscale as needed,” said Newton. “The solution needed to feature a strong architecture to support our Learning Experience Platform and our AI projects.”

Chief Operating Officer
and Senior Coach,
iCanStudy

The business determined Google Cloud was the best fit for its needs. “Google Cloud provided the easiest management and maintenance, and the partner we selected to help deliver the project was extremely comfortable with its selection,” said Newton. “Furthermore, when we started talking to the Google Cloud team, we realized its people and services could support us across the entire process, which was quite complex and required a lot of thought. In fact, Google Cloud could provide us with far more support than we had ever experienced before.”



Saving time otherwise spent onboarding developers

Without an internal development team, iCanStudy needed external expertise to help migrate to and create an architecture on Google Cloud. Based on a recommendation from the cloud provider’s Melbourne team, the education provider engaged Canberra-based specialist AI consultancy [addaxis.ai](#) to help.

addaxis.ai provided extensive support during the migration and built the architecture on which the Learning Experience Platform runs. The partner is also leading the development of iCanStudy’s AI products.

“Working with addaxis.ai has saved us the time we would

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have had to invest onboarding developers ourselves,” said Newton. “The team has handled all project management, development, and engineering tasks and provided access to expertise we simply did not have before.”

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Archer Newton
Chief Operating Officer
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iCanStudy

iCanStudy and addaxis.ai began working with Google Cloud in May 2023 on elements of the architecture for its new platform. By September the same year, iCanStudy had completed this work and in December that same year, publicly released the platform running on Google Cloud. The business has also developed two key AI products for iCanStudy customers: a Deep Search Engine and a generative-AI-based Meta Analyser. iCanStudy’s Deep Search engine enables users to write questions that aim to improve their cognitive skills and, based on program progress, provides an answer customized to their stage of skill development. For example, learners can ask cognitive questions like "My astrophysics exam is in three days and it has lots of procedural knowledge. How should I revise?" To emotional questions like, "How do I adopt a growth mindset when I always self-sabotage myself?" The answer will be personalized to their current skill level, while recommending specific resources to view next.

As a world's first, the Meta Analyser evaluates a learner's cognitive learning processes and diagnoses issues based on 100 identified patterns, which completely revolutionizes how to train one's ability to learn. Depending on the thousands variables considered in its analysis, users can submit a photo of their mind map notes or a reflection on an experience, and the Meta Analyser will provide step-by-step instructions on how to improve their next learning session, essentially acting as a coach who can see how your mind is thinking and address the areas for improvement.

The business undertook AI development, testing and prototyping in [Vertex AI](#), while the [Gemini](#) large language model powers its proprietary learning products.

The learning platform is run by a resilient, scalable cloud-based architecture, comprising services such as [Firestore](#) data storage, [VPC](#) networking, [Cloud Run](#) serverless containerization, and [Compute Engine](#) virtual machines.

“We have become the first edtech business in Australia to build on Google Cloud using generative AI through Gemini and Vertex AI,” explained Tsai.



Reducing downtime and improving reliability

Moving to Google Cloud has reduced downtime to minimal levels, while reliability has risen considerably.

“With Google Cloud, we have full control over the ability to customize the applications we can run on our platform, delivering the flexibility we need,” explained Newton. “We have also been able to create proper testing, staging, and production environments so we can undertake better, iterative testing with our customers.

“In addition, our security and compliance is more robust, since we are building on top of Google Cloud’s infrastructure with support from addaxis.ai. Finally, Google Cloud monitoring tools have made it easier for us to proactively monitor, troubleshoot, and optimize our infrastructure and applications.”

“Our partnership with Google Cloud aligns perfectly with Google's mission to make data accessible and useful for everyone. As we continue to advance our Learning Experience Platform and AI capabilities, our goals are to harness this wealth of data to revolutionize education and empower learners globally.”

Michael Tsai

With Google Cloud, iCanStudy is now well positioned to achieve its target of more than one million active users, a 200X increase from its previous capacity. “The infrastructure provided by Google Cloud enables us to scale as we need to,” said Newton.

The AI-powered products developed in Vertex AI have eliminated the need for iCanStudy to invest in 1,000-plus hours training (human) coaches before they can teach students. Additionally, user feedback is received 288X faster than before, enabling the business to respond quickly to any perceived issues. As well as increased uptime, CPU utilization has fallen from about 94% on average to just 16%, accelerating load times and improving the online user experience.

“Our partnership with Google Cloud aligns perfectly with Google's mission to make data accessible and useful for everyone,” said Tsai. “As we continue to advance our Learning Experience Platform and AI capabilities, our goals are to harness this wealth of data to revolutionize education and empower learners globally.”

iCanStudy is the world's first edtech company to combine research-backed cognitive retraining with an e-learning system and AI tools. iCanStudy has helped over 18k users achieve improvements in their learning quality and efficiency.

Industry: Education

Location: Australia

Products: [Google Cloud](#), [Cloud Run](#), [Compute Engine](#), [Firestore](#), [Gemini](#), [Vertex AI](#), [Virtual Private Cloud](#)

About Google Cloud partner- addaxis.ai

Headquartered in Canberra, Australia, **addaxis.ai** helps organizations implement AI, machine learning, and cloud to deliver value.

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Modern Infrastructure Cloud	See all products	Application modernization	Google Cloud quickstarts	Become a Partner
Multicloud		Smart analytics		Events
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Customers and case studies		Security	Learn about cloud computing	Developer Center
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